SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO2: *Experiment search and game-solving techniques such as Greedy Search, A*, CSP, Minimax, and Alpha-Beta pruning with heuristic functions for solving complex AI problems efficiently. (BL3)*

Assessment Tool Weightage: 35%

Duration: 1 Hour

QUESTIONS: 5 **Total Marks:50**

Annexure A

Scenario Based Assignment

Code of Conduct:

I RAHIMA BANU N-192525009 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:RAHIMA BANU N

Annexure A

Scenario Based Assignment

1. Scenario : A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. The environment is highly dynamic—roads may suddenly become inaccessible, and weather conditions affect travel cost. The drone has access to: A heuristic estimate (straight-line distance), Partial real-time updates about blocked paths and Variable movement costs depending on terrain and weather. However, due to urgency, the drone must quickly decide routes with minimum delay and risk.

Tasks:

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty

ii. Explain how A* would handle the same situation, including how it balances cost and heuristic

iii. Analyze the impact of heuristic accuracy on both algorithms

iv. Discuss how dynamic changes (blocked paths) affect search performance

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety

2. **Scenario**: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize Total waiting time, Fuel consumption and Traffic congestion. The system must continuously adjust signals based on traffic flow. However the search space is extremely large, traffic patterns change dynamically and the system may get stuck in locally optimal solutions

Tasks. Formulate this as an optimization problem and explain why traditional search is inefficient

i. Explain how Hill Climbing would work in this scenario and identify its limitations

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations

iv. Recommend the best approach and justify your choice based on scalability and adaptability

3. **Scenario:** An autonomous rover is deployed on Mars to explore unknown terrain. It must: navigate safely to collect samples, avoid hazards (craters, rocks), operate with limited sensing capability and make decisions without a complete map. The rover updates its knowledge as it explores, making this a real-time decision-making problem.

Tasks:

i. Explain why this problem requires an online search agent instead of offline search

ii. Analyze the challenges of partial observability and dynamic environments

iii. Describe how the rover builds and updates its internal model

iv. Compare online search with uninformed and informed search strategies

v. Justify the most suitable approach considering uncertainty, adaptability, and safety

4. **Scenario:** A university is designing an AI system to automatically generate exam timetables. The system must satisfy multiple constraints: No student should have overlapping exams, Limited number of exam halls, Certain subjects must be scheduled before others, Faculty availability constraints, Special accommodations for some students, The complexity increases as the number of courses and students grows.

Tasks :

Formulate the problem as a Constraint Satisfaction Problem (CSP)

i. Clearly define variables, domains, and constraints with explanation

ii. Explain how backtracking search works in this scenario

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency

iv. Analyze real-world challenges and justify the best strategy for scalability

5. **Scenario:** Strategic Game AI for Real-Time Decision Making (Minimax & Alpha-Beta Pruning) A gaming company is developing an AI opponent for a real-time strategy game. The AI must complete against human players, Make decisions within strict time limits, Evaluate complex game states with multiple possible moves, The decision tree is extremely large, making computation expensive.

Tasks:

i. Explain how the Minimax algorithm models this problem

ii. Analyze the computational challenges of large game trees

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning

iv. Design an evaluation function and justify the factors considered

v. Recommend a strategy for balancing optimality and real-time performance

SOLUTIONS:

1. AI-Powered Drone Delivery in a Flood-Affected Region

Scenario: A government deploys AI-powered drones to deliver emergency medicines in a flood-affected region. Roads may suddenly become inaccessible, and weather conditions affect travel costs. The drone uses straight-line distance as a heuristic, receives partial updates about blocked paths, and must quickly decide routes with minimum delay and risk.

i. Explain how Greedy Best-First Search would behave in this scenario and discuss its strengths and weaknesses under uncertainty.

Greedy Best-First Search selects the path that appears closest to the destination using only the heuristic value h(n). It makes quick decisions and reduces search time. However, it ignores the actual cost already travelled and may select a blocked or risky route.

ii. Explain how A would handle the same situation, including how it balances cost and heuristic.*

A* Search considers both the actual path cost and estimated remaining distance using:

\[f(n)=g(n)+h(n) \]

Here, g(n) represents the cost travelled, and h(n) represents the estimated distance to the destination. It selects a route with a lower total estimated cost and can find an optimal route when the heuristic is admissible and the costs are correctly represented.

iii. Analyze the impact of heuristic accuracy on both algorithms.

An accurate heuristic helps both algorithms find better routes. An inaccurate heuristic may lead Greedy Search toward a poor route. A* can still guarantee optimality with an admissible heuristic, even if the estimate is not very accurate, but it may expand more nodes.

iv. Discuss how dynamic changes (blocked paths) affect search performance.

Blocked paths force the drone to change its route. The search algorithm must update its map and recalculate the path. Frequent changes may increase computation time and delay delivery.

v. Recommend the most suitable algorithm and justify your decision considering real-time constraints, optimality, and safety.

A* Search is more suitable because it considers both actual travel cost and estimated distance. It helps identify safer and shorter routes. For real-time use, it can be combined with dynamic replanning methods to respond quickly to blocked paths and changing weather.

2. Smart City Traffic Signal Optimization

Scenario: A smart city implements an AI system to optimize traffic signal timings across hundreds of intersections. The objective is to minimize total waiting time, fuel consumption, and traffic congestion. The system continuously adjusts signals based on traffic flow.

Formulate this as an optimization problem and explain why traditional search is inefficient.

The problem can be formulated as an optimization problem where the objective is to find the best traffic signal timings. The system must minimize waiting time, fuel consumption, and congestion. Traditional search is inefficient because the search space is very large and traffic conditions change continuously.

i. Explain how Hill Climbing would work in this scenario and identify its limitations.

Hill Climbing starts with an initial signal timing and makes small changes to improve traffic flow. It selects a better neighbouring solution at each step. However, it may get stuck in a local optimum and fail to find the best overall solution.

ii. Discuss issues like local maxima, plateaus, and ridges with real-world interpretation.

- Local Maxima: The system finds a timing that improves one intersection but does not improve the entire city.
- Plateaus: Several signal timings produce almost the same traffic flow, making it difficult to choose a better solution.
- Ridges: The best solution requires changing multiple signal timings together, but changing only one at a time may not improve the result.

iii. Explain how Simulated Annealing or Genetic Algorithms can overcome these limitations.

Simulated Annealing sometimes accepts worse solutions to escape local optima. Genetic Algorithms generate multiple solutions and improve them using selection, crossover, and mutation. These methods explore a wider search space and can find better solutions.

iv. Recommend the best approach and justify your choice based on scalability and adaptability.

Genetic Algorithms are suitable because they evaluate multiple solutions simultaneously and can optimize signal timings across many intersections. They can adapt to changing traffic patterns by periodically generating new solutions.

3. Autonomous Rover Exploring Mars

Scenario: An autonomous rover is deployed on Mars to explore unknown terrain. It must navigate safely, collect samples, avoid hazards such as craters and rocks, operate with limited sensing capability, and make decisions without a complete map.

i. Explain why this problem requires an online search agent instead of offline search.

An Online Search Agent is suitable because the rover does not have complete information about its environment. It makes decisions while moving and updates its knowledge whenever it discovers new obstacles or paths. Offline search requires prior knowledge of the environment.

ii. Analyze the challenges of partial observability and dynamic environments.

The rover cannot observe the entire terrain at once because of limited sensors. Hidden rocks, craters, and unknown paths may create risks. These conditions make navigation difficult and require continuous observation and decision-making.

iii. Describe how the rover builds and updates its internal model.

The rover uses cameras and sensors to observe its surroundings. It stores information about safe paths, obstacles, and unexplored areas in an internal map. Whenever new information is received, it updates the map and selects a suitable route.

iv. Compare online search with uninformed and informed search strategies.

- Online Search: Makes decisions while exploring and updates knowledge continuously.
- Uninformed Search: Searches without using heuristic information about the goal.
- Informed Search: Uses heuristic information to guide the search toward the goal.

Online search is more suitable when the environment is unknown because it learns during exploration.

v. Justify the most suitable approach considering uncertainty, adaptability, and safety.

An Online Search Agent is the most suitable approach because it can explore unknown environments, update its internal map, and respond to newly discovered hazards. It improves adaptability and supports safer navigation.

4. AI-Based University Exam Timetable Generation

Scenario: A university is designing an AI system to automatically generate exam timetables. It must avoid overlapping exams, manage limited exam halls, consider subject ordering, faculty availability, and special student accommodations.

Formulate the problem as a Constraint Satisfaction Problem (CSP).

The exam timetable problem can be represented as a Constraint Satisfaction Problem (CSP), where variables represent exams, domains represent available time slots and halls, and constraints represent scheduling rules. The objective is to find a timetable that satisfies all constraints.

i. Clearly define variables, domains, and constraints with explanation.

- Variables: Different subjects or examinations to be scheduled.
- Domains: Available exam time slots and examination halls.
- Constraints: No student should have overlapping exams, halls must have sufficient capacity, faculty must be available, and special accommodations must be provided.

ii. Explain how backtracking search works in this scenario.

Backtracking Search assigns a time slot and hall to each examination one by one. If an assignment violates any constraint, the system goes back to the previous assignment and tries another available option. This continues until a valid timetable is generated.

iii. Discuss how constraint propagation (e.g., forward checking) improves efficiency.

Forward Checking checks the remaining possible assignments after each selection. If a time slot becomes invalid for another examination, it removes that option from the available domain. This reduces unnecessary searching and detects conflicts early.

iv. Analyze real-world challenges and justify the best strategy for scalability.

The main challenges include a large number of students, limited halls, faculty availability, and multiple scheduling conflicts. A combination of Backtracking Search, Forward Checking, and the Minimum Remaining Values (MRV) heuristic is suitable. It reduces unnecessary searches and improves timetable generation efficiency.

5. Strategic Game AI for Real-Time Decision Making (Minimax and Alpha-Beta Pruning)

Scenario: A gaming company develops an AI opponent for a real-time strategy game. The AI competes against human players, makes decisions within strict time limits, evaluates complex game states, and handles extremely large decision trees.

i. Explain how the Minimax algorithm models this problem.

Minimax is an adversarial search algorithm that considers two players: MAX and MIN. MAX tries to maximize its score, while MIN tries to minimize it. The algorithm examines possible moves and opponent responses to select the best move.

ii. Analyze the computational challenges of large game trees.

Large game trees contain many possible moves and combinations. Evaluating every possible state requires significant time and memory. This becomes difficult when the AI must make decisions within strict time limits.

iii. Explain how Alpha-Beta pruning improves efficiency with detailed reasoning.

Alpha-Beta Pruning improves Minimax by eliminating branches that cannot affect the final decision.

- Alpha (α): Best value currently available to MAX.
- Beta (β): Best value currently available to MIN.
- When $\alpha \geq \beta$, the remaining branches can be pruned.

This reduces the number of nodes evaluated and allows the AI to search deeper in less time without changing the optimal Minimax result.

iv. Design an evaluation function and justify the factors considered.

An evaluation function estimates how good a game state is for a player. A possible evaluation function is:

\[E(s)=3M+2R+2T+P \]

Where:

- M = Military strength
- R = Available resources
- T = Territory controlled
- P = Player progress

Higher scores represent better positions for MAX. These factors help the AI compare different game states and select better moves.

v. Recommend a strategy for balancing optimality and real-time performance.

A combination of Minimax, Alpha-Beta Pruning, and depth-limited search is recommended. Alpha-Beta Pruning reduces unnecessary node evaluations, while depth limits ensure decisions are made within the available time. Iterative deepening can also be used to improve decision quality gradually.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand